

Waterborne polysiloxane–urethane–urea for potential marine coatings

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Abstract Waterborne polysiloxane–urethane–ureas (WBPSUU) were prepared through a prepolymer process using siloxane polyol, namely polydimethylsiloxane (PDMS) and polyether polyol, namely poly(tetramethyleneoxide glycol) (PTMG) as the soft segments. The goal of this study was to explore the potential use of polysiloxane–urethane–urea in marine coatings in order to boost the flexibility, adhesion, erosion, and foul-release property with respect to PDMS/PTMG ratio (PDMS wt%). The structural elucidation of the synthesized waterborne polyurethane and WBPSUU was carried out by FTIR, ^1H NMR, and ^{29}Si NMR spectroscopic techniques. The XPS and AFM analyses indicate that the polymer surface can be silicone enriched with the optimum PDMS content (15.76 wt%). The antifouling property of the coatings was investigated by the immersion test under a marine environment for 90 days. The fouled area was calculated for all the samples and the fouled area (%) decreased with increasing PDMS content. After 90 days the lowest fouled area (5%) was observed in the sample using WBPSUU4 (PDMS 15.76 wt%) among all of the samples.

Keywords Polyurethane, Marine coating, Fouling, Siloxane, Immersion

Introduction

Marine biological fouling, usually termed marine biofouling, can be defined as the undesirable accumulation of microorganisms, plants, and animals on

artificial surfaces that are exposed to the marine environment. This phenomenon is clearly observed on every suitable substrate such as ship hulls, marine cages, pipelines, heat exchangers, and other structures including offshore platforms and bridges which are immersed in seawater.¹

Various antifouling paints have been developed in order to reduce the marine fouling. Among all of the different solutions that have been proposed throughout the history of navigation, tributyltin (TBT) self-polishing antifouling coatings have been the most successful in combating biofouling.^{1,2} These systems are usually acrylic resins with TBT side chains, which are released in the seawater through hydrolysis of the ester linkage, leaving a polyacrylic acid layer which is soluble in seawater and is washed off of the surface. Layer-after-layer the antifouling coating renews the surface. Thus, these systems show the same performance until the polymer film is completely washed away.^{1,3} However, the success of these paints was not long-lasting. Marine biologists have assembled the evidence of the negative effects of TBT on coastal marine life. The main reason for the adverse effects may be due to the direct dumping of organic waste containing tin during dry docking.^{2–5} The toxic compounds accumulated in the marine environment were found to be severely harmful to the non-targeted organisms.⁵ It has been shown that extremely low concentrations of TBT moiety can cause defective shell growth in the oyster and imposex. As a result, a worldwide ban on the use of TBT was implemented effectively from January 1, 2003, and the presence of such paints was banned from the surface of vessels since January 1, 2008.^{2–5}

Thus, the paint industry has focused on developing tin-free products that are able to replace the TBT-based ones. These products must yield the same economic benefits while, at the same time, must have less harmful effects on the environment. Among all of

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the possible non-toxic alternatives, one of the most promising technologies is a non-stick foul-release coating,⁴ which can prevent the adhesion of organisms by providing a low-friction, ultra-smooth surface that makes the attachment of a fouler very difficult and the detachment easy. Potential candidates for this coating include fluoropolymer and siloxane-based coatings.^{1,6,7} Ideally these “non-stick” coatings would completely prevent the attachment of foulers. However, fouling can occur on real ship panels employing non-stick coatings,^{8,9} even though these coatings significantly limit the strength of the joint between the fouler and hull, making the bond so weak that it can be broken by the weight of the fouler or the motion of the ship through the water. Therefore, more realistically, toxic antifouling paints can be replaced by non-stick coatings combined with the periodic removal of the fouling organisms under water using brushes, which is an alternative to cleaning in dry-docks.^{4,8} This combined technique could be an attractive alternative to the use of toxic antifouling materials.^{1,8,9}

Today, non-biocidal, silicone-based foul-release coatings are commercially available and minimize the adhesion of the attached organisms.^{4,9,10} These attached organisms can be removed easily when the vessel moves through the water. However, these coatings are expensive and less robust than biocidal paints, and their application is limited to fast vessels (15–30 knots) with a quick turn-around.^{9–11} Research has now focused on the development of new antiadhesive coatings that combine low adhesion properties on the “water side” to resist the attachment of the organisms in aquatic environments and high adhesion on the “substrate side” to facilitate the firm bonding to the protected object, combined with a high degree of mechanical strength.^{9–11}

Current commercial marine paints contain a large amount of volatile organic solvents,^{12,13} which are released into the atmosphere during the drying stage. These harmful solvents also threaten the human body.^{12,13} Many countries have already restricted the emission of these harmful solvents.¹³ Therefore, the marine coating must be made environmentally friendly as well as free from all kinds of pollution. Although many polymers are promising as a foul-release coating, their syntheses and applications require a large amount of organic solvent. Ideally, the polymer should be processable under moderate conditions with no (or less) volatile evolution and should be stable in a marine environment.⁴ Further, the polymer must be compatible with the additive and pigment and exhibit good foul-release properties. Only a few polymer coatings are commercially available because of an unfavorable combination of price, processability, and performance. These polymers are often made and used for specialized applications.^{3,12,13}

It has been shown that polydimethylsiloxane (PDMS) can be useful for foul-release type coatings due to its non-stick surface.¹⁴ Unfortunately, a key disadvantage of using PDMS is its poor adhesion and

bulk mechanical properties.^{14,15} Thus, over the last few years, a significant effort has been underway to overcome these limitations using other polymers such as epoxide and polyurethane with PDMS.^{14–16} However, it is hard to find the exact chemical composition of PDMS with polyurethane. As PDMS has low free energy¹⁴ and polyurethane has good elastomeric properties,¹² the exact combination of PDMS in polyurethane can be effective for antifouling properties while boosting suitable mechanical properties.

In this report, waterborne polysiloxane–urethane–ureas (WBPSUU) were synthesized by prepolymer process. Our synthesized WBPSUU coatings contain mainly water and do not emit organic solvent at any stage of synthesis or during application at drying stage. Thus, the synthesized waterborne coatings meet the environmental legislations.^{1–4,6,13} During the last decade the major improvement was to control the marine fouling by cupric oxide compounds and various organic biocides.⁴ Unfortunately, the use of cupric oxide paints might be limited in near future due to their toxicity and higher content in marine environment confirmed by recent reports.⁴ Now the promising environmentally friendly antifouling coating is the foul-release system, which has a mechanism of preventing the adhesion of foulers.^{13,14} The silicone compounds are regarded as a foul-release coating primarily as a result of their non-stick surface.⁴ Unfortunately, the silicone coating has some drawbacks such as low adhesion, less mechanical strength, brittle behavior, and high cost.¹⁴ Thus our object is to use a kind of material which can overcome these limitations. We chose polyurethane as a backbone polymer after considering advantages such as high mechanical and adhesive strength, rapid drying even at lower temperature, easy processing, and relative low cost.^{12,17–20} Since the last decade, waterborne polyurethane (WBPU) has been used in many coating and adhesive applications such as construction, automotive, packing, transportation, electronics, textiles, tape, paper, and footwear industries due to environmentally friendly technique.^{12,17–22} In this study, the ratio of poly(tetramethyleneoxide glycol) (PTMG) and PDMS was varied while other materials were fixed during preparation of WBPSUUs. WBPSUUs were synthesized using hydroxy terminated PDMS ($M_n = 550$) with PTMG ($M_n = 2000$) as a soft segment. The structure of these polyurethanes was characterized by FTIR, ¹H NMR, and ²⁹Si NMR spectroscopic techniques. The molecular weight and mechanical properties (tensile strength and Young's modulus) of the films were characterized with respect to PDMS content. The polymer surface was characterized by XPS and AFM techniques. The synthesized waterborne resin was coated onto PVC and immersed in sea water for 90 days. We checked the antifouling performance of immersed coatings after a certain interval. Photographs of the coatings were taken and the fouled area was calculated to justify the antifouling performance for all samples.

Experimental

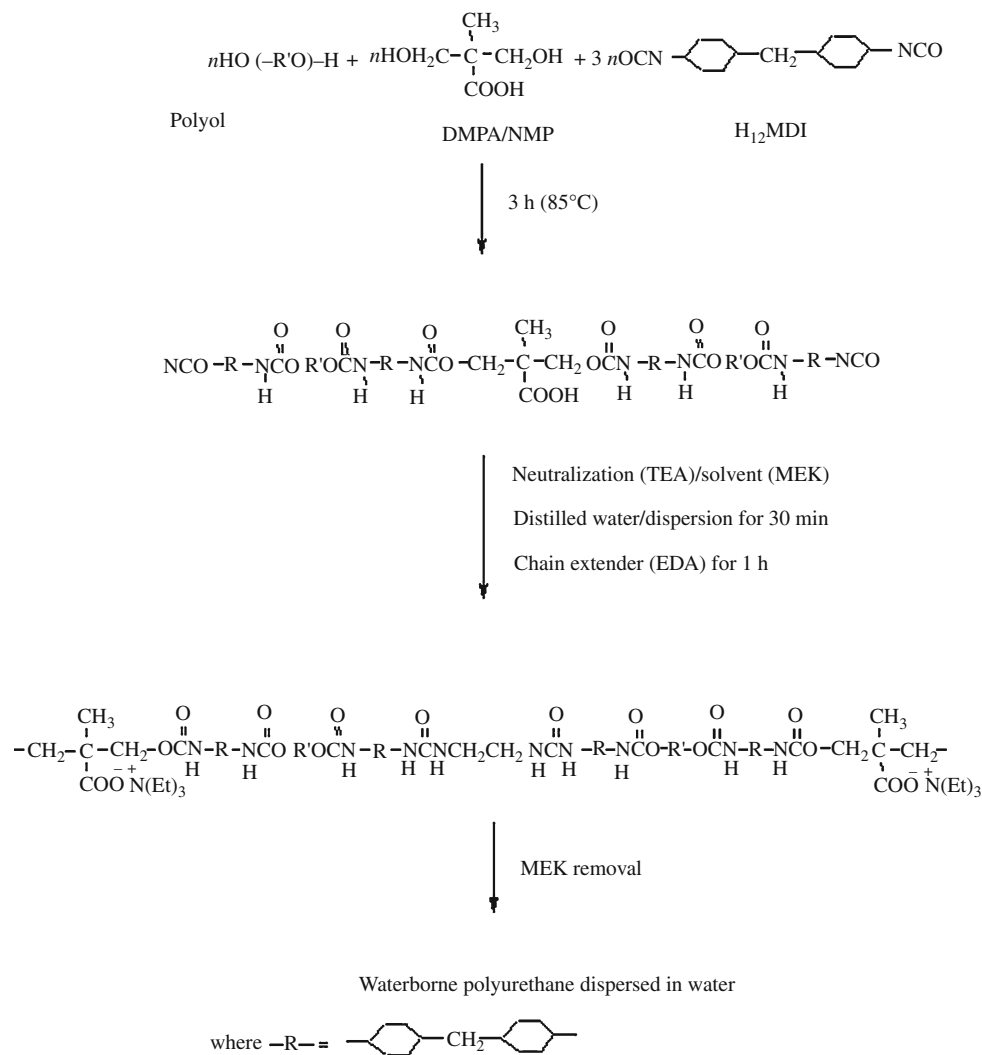
Materials

PTMG ($M_n = 2000$ g/mol, Aldrich, USA) and hydroxy-terminated PDMS ($M_n = 550$ g/mol, Aldrich) were dried under vacuum at 90°C and 1–2 mmHg for 24 h before use. Triethylamine (TEA; Junsei Chemical, Tokyo, Japan), *N*-methyl-2-pyrrolidone (NMP; Junsei Chemical), 4,4'-dicyclohexylmethane diisocyanate (H_{12} MDI, Aldrich), and ethylenediamine (EDA) (Junsei Chemical) were dehydrated with 4-Å molecular sieves for 1 week prior to use. 2,2-Bis(hydroxymethyl) propionic acid (DMPA, Aldrich) and dibutyltin dilaurate (Aldrich) were used as received.

Synthesis of WBPU and WBPSUU dispersions

WBPU and WBPSUU dispersions were synthesized using the prepolymer mixing process (Scheme 1).¹²

The polyol was placed in a four-necked flask that was equipped with a thermometer, a stirrer (Global Lab), a condenser, an inlet and outlet for dry nitrogen, and a heat jacket, and the system was degassed under vacuum at 90°C for 30 min. DMPA/NMP (1/1 w/w) was added to the mixture at 65°C and continued to be stirred for 30 min. The reaction was allowed to cool to 45°C under moderate stirring (175–200 rpm). Then dibutyltin dilaurate (0.01 wt% based on H_{12} MDI) was added to the flask along with H_{12} MDI, and the mixture was heated to 85°C under moderate stirring (175–200 rpm) and reacted for 3 h. The change in the NCO value during the reaction was determined using the standard dibutylamine back-titration method (ASTM D 1638). Then methyl ethyl ketone (MEK, 10 wt%) was added to the NCO-terminated prepolymer mixture at 65°C in order to adjust the viscosity of the solution. TEA was also added to the reaction mixture at 65°C in order to neutralize the carboxyl groups of the NCO-terminated prepolymer. After 30 min of neutralization, distilled water (70 wt%) was



Scheme 1: The preparation process for a WBPU

added to the reaction mixture (25°C) with vigorous stirring (1300–1500 rpm). The neutralized prepolymer was chain-extended by dropping EDA (mixed with water) at 40°C for 1 h, and the reaction continued until the NCO peak (2000–2300 cm⁻¹) in the IR spectrum had completely disappeared. The dispersions were obtained (30 wt% solid content) after MEK was evaporated (collected separately).

Preparation of WBPU and WBPSUU films

The films were prepared by pouring the aqueous dispersion (10 g) onto a Teflon disk (diameter 7 cm) and drying the dispersion under ambient conditions for about 48 h. The films (typically about 0.5-mm thick) were dried at 60°C for 6 h and then vacuum dried for another 12 h. The vacuum dried films were stored in a desiccator at room temperature.

Supports mounting coatings onto PVC solid supports

For the field tests, the synthesized waterborne-resins were mounted onto flat PVC solid supports through the air-spraying method. The coated samples were fully dried within 4 h at ambient temperature, and the dried samples were used in the immersion tests. Normal antifouling coatings are applied on the steel structures⁴ but in this report we used the PVC substrate because we investigated the antifouling property of coating systems excluding the corrosion phenomena by the sea water.

Field tests in seawater

The antifouling field tests were performed according to the ASTM D3623 specifications from August 2009 until November 2009 (90 days) located at Suyeong Beach in Busan, Republic of Korea. The samples that were mounted onto the PVC supports were immersed into the seawater at depths of about 1 m below the surface. The samples faced the open ocean and were not caged. Periodically, the samples were removed and photographed. We calculated the fouled area (%) using the image analysis software AxioVision Release 4.5 (Carl Zeiss).

Characterization

The mean particle size of the dispersions was measured using laser-scattering equipment (Autosizer, Malvern IIC, Malvern, Worcester, UK) at 25°C. A Fourier transform IR spectrometer (Impact 400D, Nicolet, Madison, WI, USA) was used to identify the WBPU and WBPSUU structures. The dispersion was coated on the thallium–bromide/thallium–iodide crystal surface

as a thin liquid film and dried for the analysis. For each sample, 32 scans were collected at a 4-cm⁻¹ resolution in the transmittance mode. ¹H NMR and ²⁹Si NMR spectra were obtained on a JEOL C60 HL spectrometer with CDCl₃ as a solvent. A gel permeation chromatograph (Model 500, Analytical Scientific Instruments, USA) with a refractive index detector (RI2000, Schambeck, Germany) and two Jordi gel divinyl benzene mixed bed (Jordi FLP, USA) columns were used to measure the molecular weight relative to the polystyrene standards at 30°C. The calibration curve was obtained using eight standards in the molecular weight range from 3420 × 10⁶ to 2.57 × 10⁶. The carrier solvent was tetrahydrofuran at a flow rate of 1 mL/min. The tensile properties were measured at room temperature using a United Data System tension meter (Instron SSTM-1, United Data Systems, Japan) according to the ASTM D 638 specifications. A crosshead speed of 50 mm/min was used throughout these investigations in order to determine the ultimate tensile strength, the Young's modulus, and the elongation at break (%) for all of the samples. The values are reported as the average of five measurements. The polymer surface was analyzed by XPS using an ESCA 250 X-ray Photoelectron Spectrometer (XPS) (UK) using Al K (1486.6 eV). The AFM topographs were carried out using a scanning probe microscope (SPM-Solver P47, NT-MDT, Russia) in contact mode. Both XPS and AFM analyses were done in KBSI, Pusan National University.

Results and discussion

In this report, PDMS was used with PTMG, DMPA, EDA, and H₁₂MDI and reacted to generate NCO-terminated prepolymers, followed by the processes of neutralization and dispersion to synthesize WBPSUU dispersion. It is known that the properties of WBPU depend on starting materials used and their ratio.^{1,21} The molar ratio of NCO/OH is 2.83 which was fixed for all the samples. The soft segment (polyol) content varied due to different polyol ratio of PDMS to PTMG (see Table 1). The data in Table 1 showed that all dispersions were successfully prepared. The WBPSUU6 was fully brittle at dried condition and, therefore, was not considered for characterization.

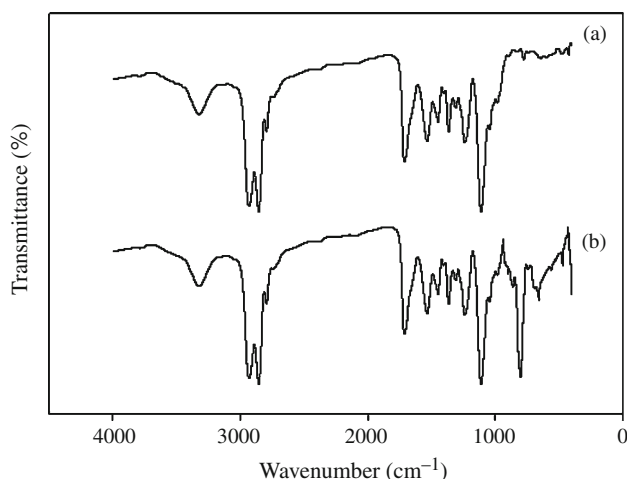
The mean particle size of all dispersions is shown in Table 2. The mean particle size varied slightly with respect to the ratio of PDMS to PTMG. The PTMG-based WBPU dispersion exhibited the smallest mean particle size among all of the dispersions. The mean particle size of any mixed polyol-based dispersions (WBPSUU1-5) was larger than that of conventional WBPU. It was also observed that the mean particle size increased with increasing PDMS content. It has been shown elsewhere²² that the macrodiol hydrophilicity can change the particle size of WBPU dispersion. The mean particle size increased with presence of

Table 1: Composition of WBPU and WBPSUU dispersions

Sample	Mole ratio							PDMS (wt%)	PTMG (wt%)	Soft segment (wt%)
	PTMG	PDMS	H ₁₂ MDI	DMPA	TEA	EDA	PDMS/PTMG			
WBPU	0.040	0	0.113	0.054	0.054	0.019	0	0	64.78	64.78
WBPSUU1	0.035	0.005	0.113	0.054	0.054	0.019	0.143	2.37	60.22	62.59
WBPSUU2	0.030	0.010	0.113	0.054	0.054	0.019	0.333	5.05	55.05	60.10
WBPSUU3	0.020	0.020	0.113	0.054	0.054	0.019	1.0	11.64	42.33	53.97
WBPSUU4	0.015	0.025	0.113	0.054	0.054	0.019	1.667	15.76	34.39	50.15
WBPSUU5	0.010	0.030	0.113	0.054	0.054	0.019	3.0	20.63	25.00	45.63
WBPSUU6	0	0.040	0.113	0.054	0.054	0.019	0	33.59	0	33.59

Table 2: Particle size of dispersions, molecular weight, mechanical properties, and RMS values of WBPU and WBPSUU films

Sample	Particle size (nm)	Molecular weight (M_n)	Young's modulus (MPa)	Tensile strength (MPa)	Elongation at break (%)	RMS roughness (nm)
WBPU	70	25000	5.0	32	698	0.681
WBPSUU1	76	23000	4.9	31	692	0.457
WBPSUU2	91	21000	4.7	29	681	0.372
WBPSUU3	105	20000	4.0	22	653	0.246
WBPSUU4	115	16000	2.0	15	350	0.148
WBPSUU5	126	12000	1.0	6	156	1.338

**Fig. 1: FTIR spectra of WBPU (a) and WBPSUU4 (b)**

hydrophobic macrodiol.²² This is well known that PDMS is more hydrophobic than PTMG,¹⁸ thus the mean particle size increased with increasing PDMS content.

The structure of conventional WBPU and WBPSUU films was identified using characteristic IR peaks (Fig. 1). The absence of a peak in the range of 2000–2300 cm⁻¹ indicated that all of the isocyanate groups reacted in this system.^{12,20,22} The spectrum is mainly characterized by the bands at 3150–3600 cm⁻¹ (NH stretching vibrations), 2800–3000 cm⁻¹ (CH stretching vibrations: antisymmetric and symmetric

stretching modes of methylene groups), 2795 cm⁻¹ (O–CH₂ stretching), 1600–1760 cm⁻¹ (amide I: C=O stretching vibrations), 1540 cm⁻¹ (amide II, $\delta_{N-H} + \nu_{C-N} + \nu_{C-C}$), 1336–1388 cm⁻¹ (ν_{C-N}), 1183–1292 cm⁻¹ (amide III, ν_{C-N}), 1109 cm⁻¹ (C–O–C stretching vibration, ether group), and 766 cm⁻¹ (amide IV). The band at 1415 cm⁻¹ is attributed to the CH₂ scissoring and the CH₃ deformation. The absorbance in between 1002 and 1012 cm⁻¹ is attributed to the stretching and rocking vibrations of the C–C and CH₂ groups, respectively. The amide I vibration consists of several components that reflected the C=O groups in different environments and is sensitive to the specificity and the magnitude of the hydrogen bonding. The amide I mode is a highly complex vibration mode that involves contributions from the C=O stretching, the C–N stretching, and the C–C–N deformation vibrations.^{17,18} The amide II mode is a mixture of the N–H in-plane bending, the C–N stretching, and the C–C stretching vibrations and is sensitive to both the chain conformation and the intermolecular hydrogen bonding. The amide III mode involves the stretching vibration of the C–N group. Amide III is highly mixed and complicated by the coupling of the NH deformation modes that are observed between 1181 and 1292 cm⁻¹.^{17–19} Amide IV, V, and VI bands are highly mixed modes containing a significant contribution from the NH out-of-plane deformation mode, and they are expected to be in the 800–400 cm⁻¹ region. A very weak single band is observed at 833 cm⁻¹, which corresponded to either the coupled vibration of the C–O stretching or the CH₂ rocking modes. The strong IR band assigned to the

asymmetric stretching vibration of the C–N group is expected at 1040 cm^{-1} . This band overlapped with the very strong band at 1109 cm^{-1} for the C–O–C stretching vibration of the ether groups in the WBPU films.^{17,18} In WBPSUU, the additional characteristic methyl peak (Si–CH₃) appears at 806 cm^{-1} and confirmed the presence of PDMS in WBPSUU.¹⁸ However, the characteristic Si peak cannot be isolated due to the ether band in the same region of WBPSUU.

¹H NMR spectroscopy provides more conclusive evidence of the structure of WBPU and WBPSUU. The ¹H NMR spectra of WBPU and WBPSUU in CDCl₃ at room temperature are shown in Figs. 2 and 3, respectively. The corresponding chemical shift assignments are listed in Table 3. There are two NH groups in both the WBPU and WBPSUU: one in the urethane group and the other in the urea group. The peak at 7.11 ppm corresponds to the urea –NH group and the

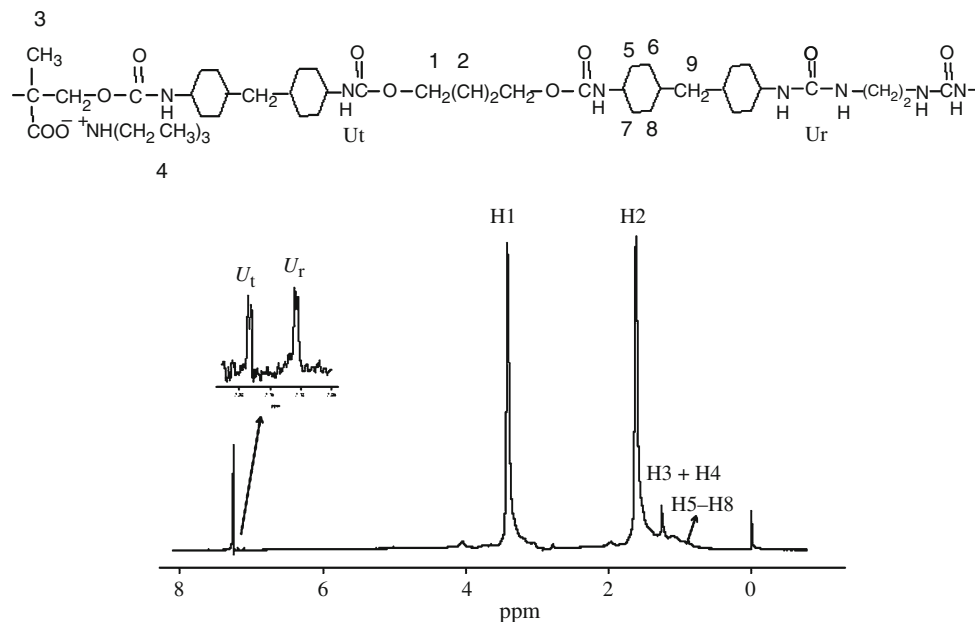


Fig. 2: ¹H NMR spectrum of WBPU in CDCl₃

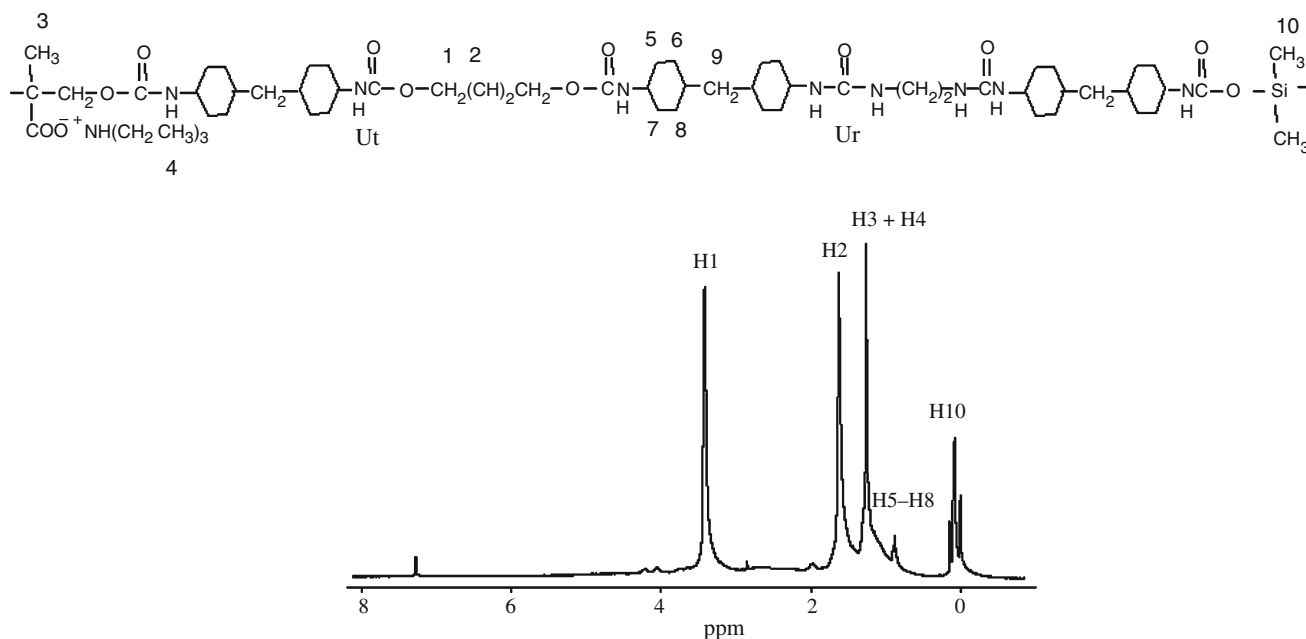


Fig. 3: ¹H NMR spectrum of WBPSUU4 in CDCl₃

peak at 7.18 ppm corresponds to the urethane –NH group. This figure shows that the hydroxyl groups were completely reacted with H₁₂MDI due to the disappearance of the OH at 4.68 ppm.^{23,24} These are consistent with the findings of previous reports^{12,20,23–25} suggesting the successful synthesis of WBPU and WBPSUU. In addition, the H₃C–Si appears at 0.3 ppm^{25,26} further confirming the chemical bonding of PDMS with H₁₂MDI in WBPSUU. The WBPSUU was further characterized with the ²⁹Si NMR spectroscopic technique. Figure 4 shows only one chemical shift value corresponding to the silicon atoms present in the WBPSUU4. The chemical shift at –112.34 ppm corresponds to the silicon atom^{26–32} present in the WBPSUU4. The presence of this chemical shift also confirms the complete reaction between PDMS and H₁₂MDI during the WBPSUU synthesis.

The molecular weight is considered to be one of the most important parameters of polyurethane. The mechanical and adhesive properties are significantly affected by the molecular weight of polyurethane.^{20,21}

Table 3: Assignment of ¹H NMR and chemical shift of WBPU

Assignment	Chemical shift (ppm)
NH(Ut)	7.18
NH(Ur)	7.10
H1	3.34
H2	1.65
H3 + H4	1.25
H5–H8	1.0–1.2

Ut, urethane; Ur, urea

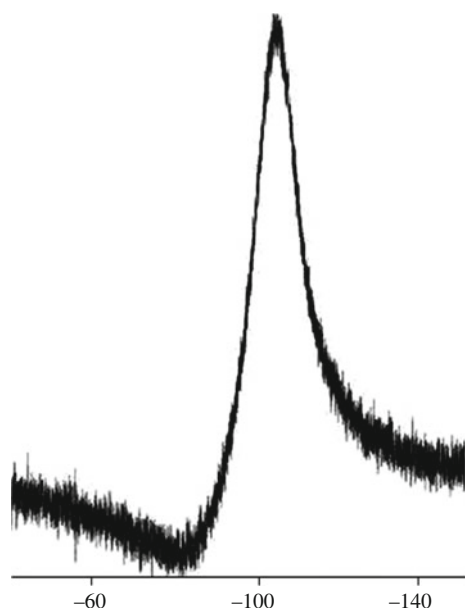


Fig. 4: ²⁹Si NMR spectrum of WBPSUU4 in CDCl₃

The polyurethane molecular weight depends on the polyol, chain extender, and diisocyanate content.^{19,22} The conventional WBPU (PTMG polyol-based) exhibited the highest molecular weight among all of the samples. The mixed polyol-based WBPSUU samples exhibited lower molecular weight than those for PTMG-based polyurethane (see Table 2). Moreover the molecular weight gradually decreased with increasing PDMS content. As we decreased the PTMG content using higher PDMS content of short chain length, the soft segment content decreased in WBPSUU and thus decreased the molecular weight.^{33–38}

The tensile strength, Young's modulus, and elongation at break (%) of all films are summarized in Table 2. All three of these properties depended on the polyol ratio. The PTMG-based WBPU exhibited the maximum tensile strength and the Young's modulus among all of the samples. The mixed polyol-based WBPSUU had lower tensile strength, Young's modulus, and elongation at break (%) compared with those for conventional WBPU. All of these properties decreased with increasing PDMS content. However, above 11.64 wt% PDMS, the properties dropped sharply because the two polyols have significantly different polarity and may promote the phase separation that decreases the mechanical properties of polymer.^{33,37} Above 20.63 wt% PDMS, the phase separation reached a critical point, where very poor interfacial adhesion occurred, and the properties rapidly decreased. Therefore, a reasonable balance between the phase separation and interfacial adhesion is needed in order to achieve good mechanical properties in the WBPSUU films.

Figure 5 shows the XPS survey scan of WBPU and WBPSUU. The peaks at 531, 285, 101, and 150 eV due to oxygen (1s), carbon (1s), silicon (1s), silicon (2p), respectively, are observed in the survey spectra of

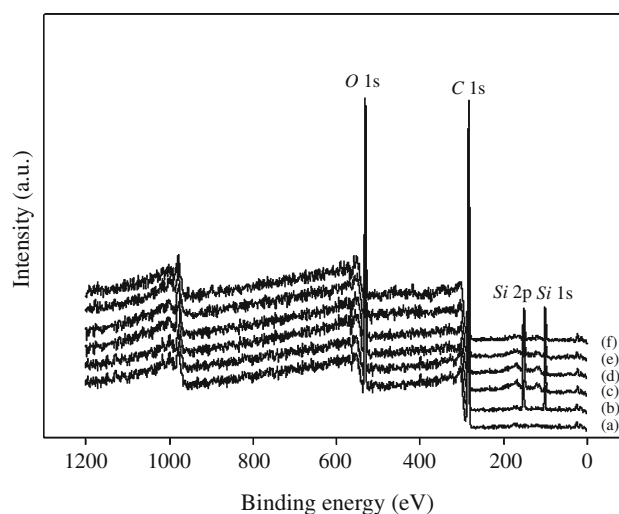


Fig. 5: XPS scan spectra of WBPU (a), WBPSUUU1 (b), WBPSUUU2 (c), WBPSUUU3 (d), WBPSUUU4 (e), and WBPSUUU5 (f)

WBPSUU, whereas the peaks at 531, 285, and 400 eV (nitrogen 1p) are observed for conventional WBP. It is well established that the chemical nature of the surface is different from the bulk composition in polyurethane.³³ Since the hard segments of polyurethanes are usually in their glassy state, the soft segments have higher mobility and can easily migrate to the surface region. Thus, the structures of the polyurethanes are quite complicated and associated with a number of carbon atoms in different environments, and are related to their different binding energies. Again C 1s binding energy of these functional groups is very close to each other and difficult to resolve in the curve fitting analysis. Therefore, these groups are combined and classified into four or five component groups (the deconvoluted spectra is shown in Fig. 6) corresponding to the C–Si carbon atom appearing at 280.8–283.8 eV (except WBP film), the C=O carbon atom appearing at 289.7–285.8 eV (except

WBPSUU4 and WBPSUU5 film), a peak associated with the C–C or C–H at 282.0–285.9 eV and the C–O and C–N contributions are observed in the range 284.1–286.1 and 284.5–287.1 eV, respectively. The relative area of each group is presented in Table 4. It is noteworthy that the carbonyl contribution attributed to the urethane/urea groups and the peak area decreased with increasing PDMS content. Most importantly the C=O peaks are absent in WBPSUU4 and WBPSUU5 samples. This indicates that the polar carbonyl group is buried in the bulk of the sample. In addition, the peak for C–Si appears strongly for these two samples. This suggests that there is a higher concentration of silicone on surface for two samples WBPSUU4 and WBPSUU5. This can be ascribed to the fact that PDMS segment tends to segregate and remain in the air–polymer interface because of its lower surface energy compared to bulk in polyurethane.^{14,39}

AFM is a widely used technique to determine the surface topography of polymer films. Figure 7 shows the typical AFM topography of WBP and the WBPSUU4. The root mean square (RMS) roughness values are summarized in Table 2. The RMS value decreased with increasing PDMS content up to 15.76 wt%. The RMS values and images of all samples suggest that the WBPSUU4 film is relatively smoother. This correlates well with the XPS findings that silicone

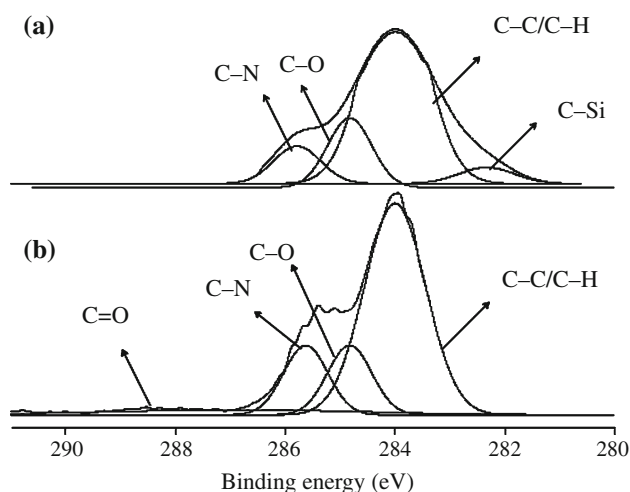


Fig. 6: Curve fitting of XPS C 1s core level spectra of WBPSUU4 (a) and WBP (b)

Table 4: Relative peak area (%) contribution of various carbon functional groups evaluated from XPS

Sample	C–H/C–C	C–N	C–O	C=O	C–Si
WBP	62.5	16.0	15.0	6.5	...
WBPSUU1	63.5	14.5	15.5	6.0	0.5
WBPSUU2	65.0	12.5	16.0	5.0	1.5
WBPSUU3	66.5	10.5	17.0	3.0	3.0
WBPSUU4	69.0	8.0	18.0	0	5.0
WBPSUU5	70.0	6.0	18.5	0	5.5

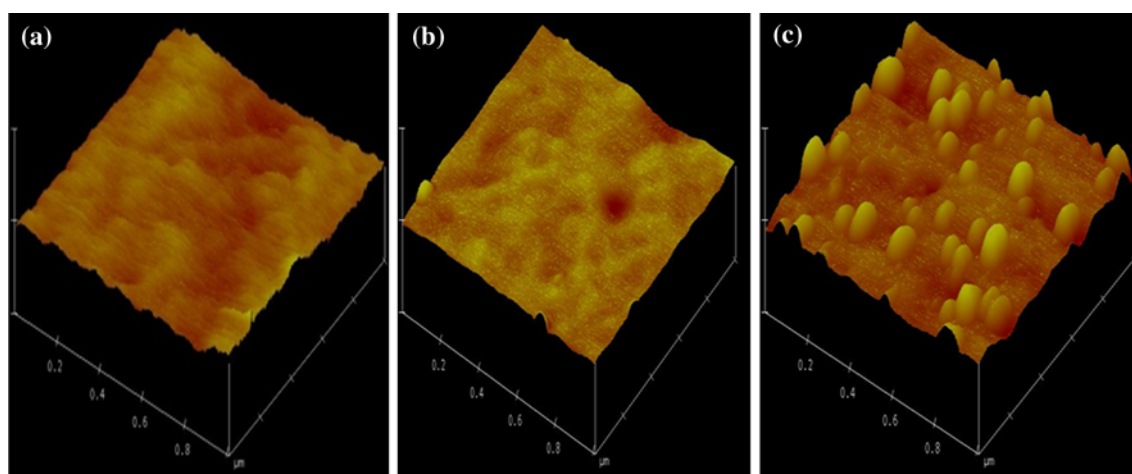


Fig. 7: AFM 3D topographs of WBP (a), WBPSUU4 (b), and WBPSUU5 (c)

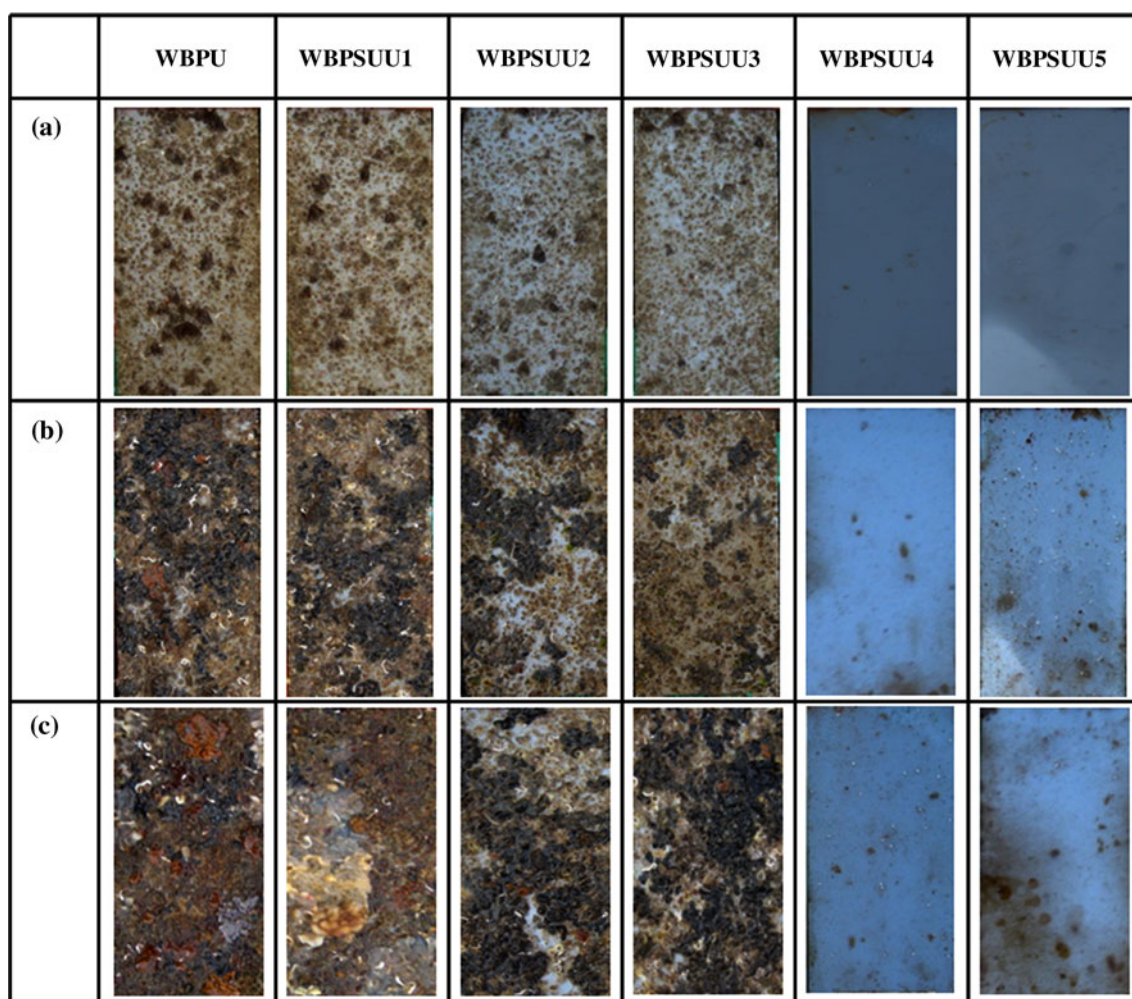


Fig. 8: Photographs of all samples after immersion in sea water for 30 days (a), 60 days (b), and 90 days (c)

moieties enrich the surface of the coatings, rendering it more planar and smooth. However, the RMS value is slightly higher in WBPSUU5. This can be ascribed to the fact that the above optimum silicone (PDMS) content has a tendency to aggregate^{40,41} and thus increase the RMS value slightly.

Seawater-immersion test of pure WBPU and WBPSUU coatings were carried out to confirm the antifouling property depending on the PDMS contents. The antifouling, erosion, and adhesion properties were examined by visual inspection. The fouled area (%) was also calculated using the image analysis software AxioVision Release 4.5 (Carl Zeiss). The pictures corresponding to 90 days of immersion are presented in Fig. 8, and all of the coatings were free from erosion. However, the coating using WBPSUU5 (PDMS 20.63 wt%) started to slightly pull away from the PVC-surface at third month. This might be due to the lower adhesive strength when using higher PDMS content.⁴² The coating using WBPU was covered with various marine foulers including juvenile barnacles, oysters, polychaetes, and a thin slime of an algal mat.

Heavy macrofouling growth was also observed on this coating. On the other hand, the coatings using WBPSUUs exhibited a decrease growth rate of marine fouling as the PDMS content increased. The lowest (almost free from fouling) marine fouling growth rate on coating was observed using 15.76 wt% PDMS (WBPSUU4). Above 15.76 wt% PDMS, the marine fouling growth rate was almost constant. Unfortunately, the coating lost some mechanical strength at this stage and thus the coating began to pull away from the PVC-surface. The WBPSUUI contains lower PDMS content (2.37 wt%) and the surface is slightly silicone enriched; hence, this sample almost failed to combat against biofouling (see Fig. 8). However, with increasing PDMS content, lower biofouling was observed.^{14,39} The WBPSUU4 and WBPSUU5 coatings were highly silicone enriched (confirmed by XPS and AFM) and made the coatings very smooth and hence the samples were almost free from biofouling. The fouled area (%) of all coatings is summarized in Table 5. This value is almost 99% for WBPU after 90 days. In WBPSUU coatings, the fouled area (%)

Table 5: Fouled area of coated samples after immersion in sea water using WBPU and WBPSUU dispersions

Immersion days	Fouled area (%)					
	WBPU	WBPSUU1	WBPSUU2	WBPSUU3	WBPSUU4	WBPSUU5
30	78	77	76	73	2	2
60	96	88	85	81	3	4
90	99	97	91	88	5	6

decreased with inclusion of PDMS. Initially, the fouled area decreased slightly in WBPSUU2 and WBPSUU3 coatings. The fouled area (%) decreased dramatically for WBPSUU4 and WBPSUU5 coatings. This implies that after certain PDMS content is reached, the attachment of fouler on coating was very difficult due to very smooth surface and it acted as a foul-release coating. In this study, the coatings were prepared using two different polyols. In WBPSUU, the PDMS remained on the surface (contact with water) and the adhesive strength between the coating and PVC was controlled by the PTMG and hard segment. The dual mechanism of this synthesized WBPSUU made the resin very suitable for marine coating. The commercial marine paint mainly consists of polymer resins, pigments, additives, and organic solvents. The examined coating consists of the synthesized waterborne polymer resins only. The WBPSUU with optimum PDMS content (15.76 wt%) can be more effective with suitable pigments and additives to combat the marine fouling. The synthesized polymer resin mainly dispersed in water. When it will be coated on a ship hull only water will evaporate and can save the atmosphere from pollution. Thus, the system is fully environmentally friendly.¹³ Investigations are ongoing in order to determine the long-term immersion behavior in sea water. The properties of the WBPSUU resin can be adjusted by changing the various reactants and their ratio.

Conclusions

WBPSUU resins were successfully prepared with various compositions of polyols PDMS and PTMG. By XPS and AFM analyses it was confirmed that the surface of the coating was silicone enriched in PDMS at a content of 15.76 wt% and thus made the coating very smooth which can prevent marine fouling after immersion in sea water. The adequate balance of PDMS and PTMG kept the mechanical strength and adhesive strength of the WBPSUU coating adequate after 90 days of water exposure. This study also confirmed that smoothness can be the result of more than chemical composition. A surface enriched in PDMS is not responsible per se for a smooth coating or the only reason for the observed good antifouling performance. It is more likely that the reason for good performance is because of other properties such as a combination of a low surface energy (caused by

PDMS) and low modulus. It was observed that the conventional WBPU-coating exhibited marine fouling after immersion in seawater. The decrease in fouled area (%) implied that the growth of marine fouling decreased with increasing PDMS content on WBPSUU coatings. After 90 days the fouled area was 99 and 5% for WBPU and WBPSUU4, respectively. Thus, the synthesized WBPSUU4 can be a good candidate for resolving the current scarcity of environmentally friendly marine coating.

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